Lecture 09 — Nearest neighbors

Wed Sep 16

Last updated: September 13, 2026.

Boyd video (optional): Lecture 10 — distance / nearest neighbors · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd §3.2 nearest neighbor; applications in §3.2 and exercises 3.10–3.11, 3.24. (Course hub lists Boyd 4.1 for this slot historically; nearest-neighbor ideas live in Chapter 3.) Related objectives: Lectures. Schedule. Quiz 3 is Friday, September 18.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Check yourself**.

Learning objectives

By the end of class, students should be able to:

1. Formulate a nearest-neighbor question using vectors.
2. Find a nearest neighbor in a small dataset by hand.
3. Explain how feature scale can affect the answer.
4. Justify a choice of distance or similarity measure.

Warm-up

Patients as 2-vectors (fever °F above 98.6, cough score 0–10):

$$p = (1.2,\ 3), \quad q = (0.1,\ 8), \quad r = (1.0,\ 4).$$

A new patient is $x = (1.1,\ 3.5)$. Which of p, q, r is nearest to x in Euclidean distance? Compute the three distances.

Fill in

Given candidates $z_1, \dots, z_m$ and a query x, a **(distance) nearest neighbor** is any z_j with

$$\|x - z_j\| \le \|x - z_i\| \quad \text{for all } i.$$

In words: the candidate with _____ Euclidean distance to x.

An **angle nearest neighbor** minimizes $\angle(x, z_i)$ instead (equivalently maximizes cosine similarity when angles are in $[0, \pi]$).

Feature vectors often mix units (age, dollars, counts). Then $\|x-y\|^2$ adds _____ of entrywise differences, so a large-scale feature can _____ the distance. Remedies include choosing units carefully, _____ features, or using a _____ (Boyd 3.28).

Choose Euclidean distance when "close in feature space" means similar _____. Choose angle/cosine when overall _____ should not matter (documents, some sensors).

Software

Octave — distances from `x` to rows of a matrix `Z` (one candidate per row):

```
x = [1.1; 3.5];
Z = [1.2, 3; 0.1, 8; 1.0, 4];   % each row a candidate
d = sqrt(sum((Z - x').^2, 2))  % 3 distances
[~, j] = min(d)                % nearest index
```

Python:

```
import numpy as np
x = np.array([1.1, 3.5])
Z = np.array([[1.2, 3.0], [0.1, 8.0], [1.0, 4.0]])
d = np.linalg.norm(Z - x, axis=1)
j = np.argmin(d)
```

Worked in class

W1. Formulate. "Among yesterday's five Arcata weather vectors, which day is most like today?" Write this as a nearest-neighbor problem: what are x and $z_1, \ldots, z_5$, and what distance will you use?

W2. Five candidates (Quiz 3 C1.1). $x = (2, 1)$ and

$$z_1 = (2, 2),\ z_2 = (0, 1),\ z_3 = (3, 1),\ z_4 = (2, 0),\ z_5 = (1.5, 1.5).$$

Compute all five distances and name the nearest neighbor.

W3. Rescale one feature (Quiz 3 C1.2). Same points, but replace the first coordinate by 10 times its value (so x becomes $(20, 1)$, etc.). Recompute nearest neighbor. Did it change?

W4. EHR story (Boyd 3.11). In one or two sentences, give a practical use for finding the 10 nearest neighbors of a patient's feature vector among other patients.

W5. Nearest nonnegative vector (Boyd 3.13 idea). For $x = (3, -2, 0.5, -4)$, what nonnegative y minimizes $\|x - y\|$? (Entrywise: you cannot do better than clipping negatives to 0.)

Check yourself

C1. $x = (0, 0)$, candidates $(1, 0)$, $(0, 2)$, $(0.5, 0.5)$. Nearest?

C2. Why might cosine similarity beat Euclidean distance for bag-of-words document vectors?

C3. After standardizing every feature (mean 0, std 1 across the dataset), what problem with heterogeneous units have you partly addressed?

C4. True or false: the nearest neighbor is always unique.

C5. Write Octave **or** Python that prints the index of the nearest row of `Z` to `x`.

Answers (Check yourself)

C1. Distances 1, 2, $\sqrt{0.5} \approx 0.707$. Nearest is $(0.5, 0.5)$.

C2. Document length (total words) varies a lot; cosine compares direction of the word pattern and ignores overall scale.

C3. Features no longer live on wildly different numeric scales, so one feature is less likely to dominate Euclidean distance.

C4. **False.** Two candidates can tie for smallest distance.

C5. Octave: `[~,j]=min(sqrt(sum((Z-x').^2,2)))`. Python: `j=np.argmin(np.linalg.norm(Z-x,axis=1))`.